SMART INDIA HACKATHON 2026

TITLE PAGE

Problem Statement ID

SIH26261

Problem Statement Title

AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification

Theme

Smart Automation

PS Category

Software

Team ID

Team Name (Registered on portal)

Pramaan

Every AICTE application verdict, with proof



Team Name

IDEA TITLE

From manual scrutiny to one verdict, with proof

SMART INDIA HACKATHON 2026

Today

- Officials read every land, building, fire and faculty document by hand
- An edited certificate looks genuine to the eye
- Nobody can compare one faculty list with every other institution's

Proposed Solution

Pramaan reads an institution's whole application, checks it against the AICTE Approval Process Handbook, against itself and against every earlier application, and returns one verdict with quoted evidence.

COMPLIANT | **REJECT** | **HUMAN REVIEW**

Rule: never approves what it is unsure of. Unreadable text, unverified AI readings and edit marks all go to an officer, with the reason.

01 Read
typed PDFs, scans and phone photos
OCR runs offline on the server

02 Check norms
faculty ratio, built-up area, fire validity
from the AICTE handbook, with annexure cited

03 Authenticate
edit marks, QR vs page, digital signatures

04 Remember
same faculty or certificate at two
institutions is caught across applications

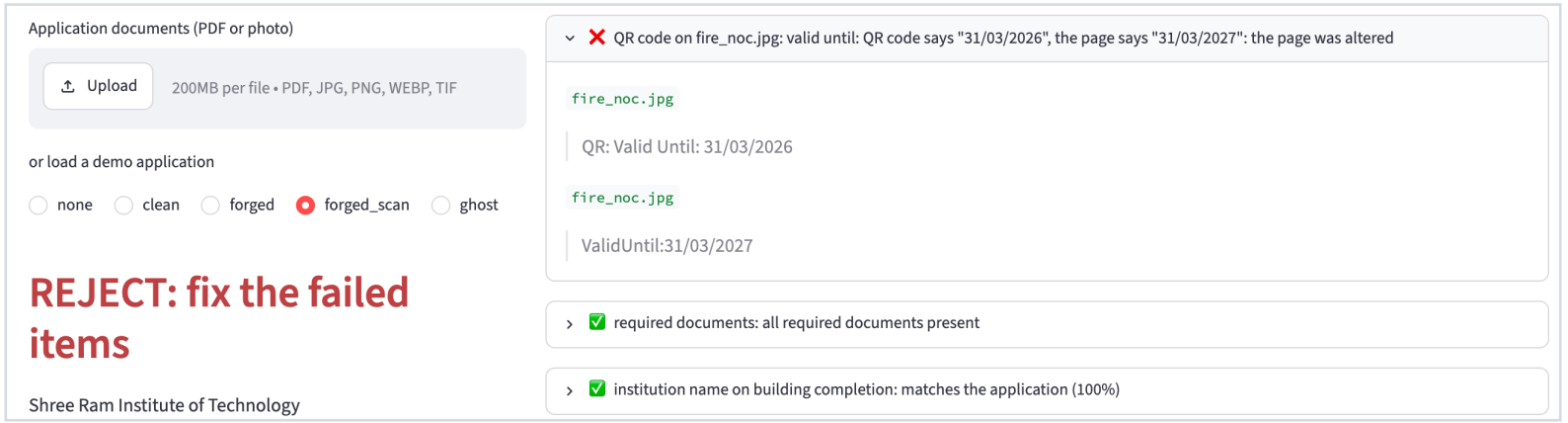
Team Name

TECHNICAL APPROACH

SMART INDIA HACKATHON 2026

AI proposes. Code verifies.

Read PyMuPDF RapidOCR → **Classify** document type → **Extract** labels, then Gemini → **Verify** quote must be in the text → **Check** norms and cross-document → **Authenticate** edits, QR, signature → **Registry** against earlier applications → **Verdict** with quoted evidence



Example: a forger prints and scans the page, so no edit marks survive. The issuer's QR code still holds the true date, so Pramaan rejects it.

Python | **FastAPI**
Streamlit | **SQLite**
PyMuPDF | **RapidOCR**
OpenCV | **pyHanko**
Gemini | **RapidFuzz**
ReportLab | **pytest**

Team Name

FEASIBILITY AND VIABILITY

Working prototype, real norms, honest limits

28
automated tests
all passing

7
document types
read

<1 s
per demo
application
on a laptop

REST
API ready for
portal integration

4 applications, riskiest first

Institution	Verdict	Failed	Review	Passed	Checked
Shree Ram Institute of Technology	REJECT: fix the failed items	3	3	9	2026-10-31 12:27:09
Ganga College of Engineering	REJECT: fix the failed items	2	0	12	2026-10-31 12:27:23
Shree Ram Institute of Technology	REJECT: fix the failed items	1	0	11	2026-10-31 12:27:21
Shree Ram Institute of Technology	COMPLIANT	0	0	12	2026-10-31 12:26:55

Viability

- Open-source stack; OCR and every rule run on the server
- Norms live in one config file, updated each handbook cycle

Risks and how we handle them

Real scans are messier than our samples

Tested on synthetic documents so far; unreadable fields go to a human

AI can invent a value

Kept only if its source line is in the document, numbers exact

A careful paste on a photo with no QR

Not caught yet; issuer QR and signatures are the defence

Sensitive documents

AI reader is optional; offline mode sends nothing out

Team Name

IMPACT AND BENEFITS

Faster scrutiny, harder fraud, decisions you can audit

SMART INDIA HACKATHON 2026

AICTE officers

- Clean files verified at once
- Riskiest applications first
- Every finding quotes its source

Honest institutions

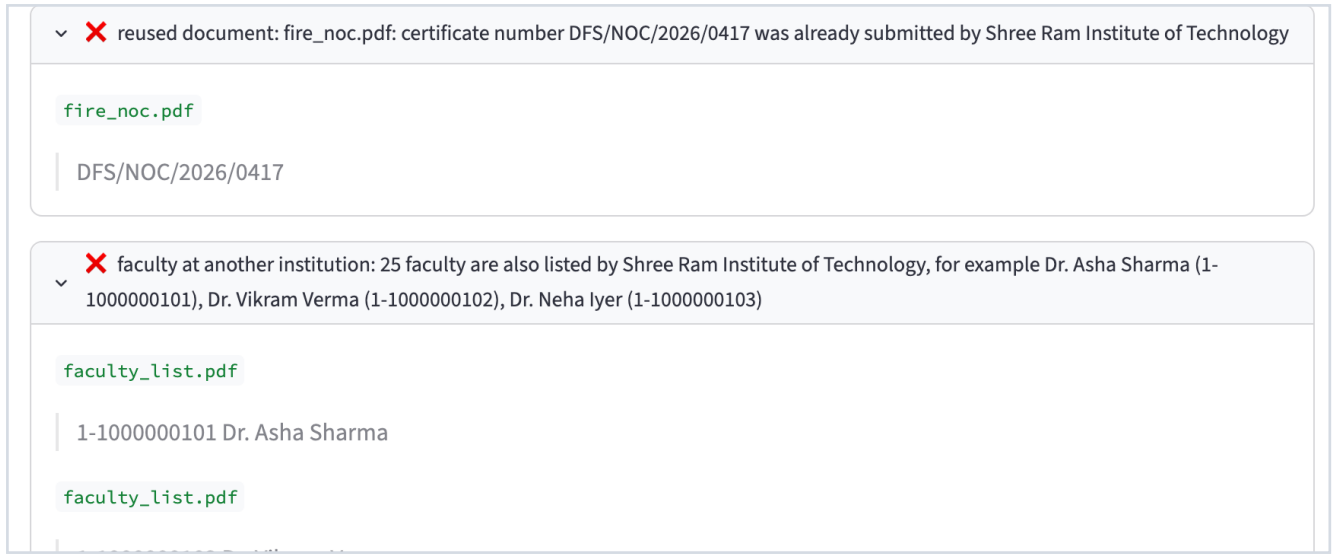
- Know exactly what to fix
- Consistent rules for everyone
- No first-glance bias

Students

- Real faculty, not names on paper
- Safer, fire-certified buildings
- Trustworthy approvals

Government

- Audit trail for every decision
- Ghost faculty detected at scale
- Portal plugs in over REST



Caught across applications

A second college lists 25 faculty already on another institution's record, and reuses its fire certificate number.

Every document looks fine alone. Only memory across applications finds it.

Team Name

RESEARCH AND REFERENCES

Built on the AICTE handbook and published standards

SMART INDIA HACKATHON 2026

Regulation

AICTE Approval Process Handbook 2024-27
Annexure 3 (land, area), Annexure 5 (faculty), clause 6.17

AICTE Act, 1987
Statutory basis for approval of technical institutions

Information Technology Act, 2000
Legal recognition of digital signatures

Digital Personal Data Protection Act, 2023
Handling of faculty and applicant data

Standards and research

ISO 32000-2
PDF structure, incremental updates, signatures

ISO/IEC 18004
QR code symbology

Du et al., 2020, PP-OCR
Lightweight OCR, arXiv:2009.09941; basis of RapidOCR

Levenshtein, 1966
Edit distance, used for name matching